Part 3

Gradients: Iterations

$$y = m_1 x_1 + m_2 x_2 + b$$

$$\hat{y} = m_1 x_1 + m_2 x_2 + b$$

$$\hat{y} = X m + b$$

$$b = [1, 1]$$

$$fmX = \begin{bmatrix} 1 & 3 \\ 4 & 10 \end{bmatrix}$$

$$vy = \begin{bmatrix} 5 \\ 6 \end{bmatrix}$$

$$vm = \begin{bmatrix} -1 \\ 2 \end{bmatrix}$$

$$b = 1$$

$$J(m,b) = \frac{1}{n} \sum_{i=1}^{n} (\hat{y}_i - y_i)^2$$

$$\frac{\partial J}{\partial m} = \frac{2}{n} X^T (\hat{y} - y)$$

$$\hat{y} = Xm + b = \begin{bmatrix} 1 & 3 \\ 4 & 10 \end{bmatrix} \begin{bmatrix} -1 \\ 2 \end{bmatrix} + \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$\hat{y}_1 = 1(-1) + 3(2) + 1 = 6$$

$$\hat{y}_2 = 4(-1) + 10(2) + 1 = 17$$

$$\hat{y} = \begin{bmatrix} 6 \\ 17 \end{bmatrix}$$

Error and Cost

$$\text{Error} = (\hat{y} - y) = \begin{bmatrix} 6 \\ 17 \end{bmatrix} - \begin{bmatrix} 5 \\ 6 \end{bmatrix} = \begin{bmatrix} 1 \\ 11 \end{bmatrix}$$

$$J = \frac{1}{2} (1^2 + 11^2) = \frac{1}{2} (1 + 121) = 61$$

Compute gradient

$$\frac{\partial J}{\partial m} = X^T(\hat{y} - y)$$

$$= \begin{bmatrix} 1 & 4 \\ 3 & 10 \end{bmatrix} \begin{bmatrix} 1 \\ 11 \end{bmatrix} = \begin{bmatrix} 1(1) + 4(11) \\ 3(1) + 10(11) \end{bmatrix}$$

$$= \begin{bmatrix} 45 \\ 113 \end{bmatrix}$$

$$\Sigma(\hat{y} - y) = 1 + 11 = 12$$

Update parameter

$$M_{new} = m - \alpha \frac{\partial J}{\partial m}$$

$$= \begin{bmatrix} -1 \\ 9 \end{bmatrix} - 0.01 \begin{bmatrix} 45 \\ 113 \end{bmatrix}$$

$$= \begin{bmatrix} -1 - 0.45 \\ 2 - 1.13 \end{bmatrix} = \begin{bmatrix} -1.45 \\ 0.87 \end{bmatrix}$$

$$b_{new} = b - \gamma \frac{dJ}{db}$$

$$= 1 - 0.01(12) = 1 - 0.12 = 0.88$$